
Straight Line Complexity of Linear Programs

The last two lectures explained why random perturbations aid the simplex method: they remove pathological behavior. We now turn to the second main family of LP algorithms, interior point methods, which unlike the simplex method solve linear programs in (weakly) polynomial time. This lecture develops a geometric-combinatorial complexity measure for linear programs, the *straight line complexity* (SLC), which captures the best iteration count that *any* path-following interior point method can achieve: it yields lower bounds for all such methods — exponential ones, for carefully constructed instances — and, for a suitable method presented in Lecture 5, approximately matching upper bounds. Circuits, introduced in Lecture 3, reappear as the elementary building blocks controlling this measure.

1 Interior Point Methods, and the Iteration Question

Recall the division of LP algorithms from Lecture 2 into exact methods, such as the simplex method, and approximate methods, which drive down the optimality gap and whose natural complexity question is: *how many iterations to halve the gap?* Interior point methods are the leading approximate solvers. They work by following the *central path*, a trajectory through the interior of the feasible region that trades off staying as far from the boundary as possible against making progress toward the optimum (Section 2 makes this precise), and for them the classical answers are well understood. The log-barrier path-following method of Renegar [Ren88] halves the gap every $O(\sqrt{n})$ iterations, each iteration solving one linear system; over the rationals, running it for $O(\sqrt{n}L)$ iterations — L being the bit size of the input — shrinks the gap below $2^{-\Theta(L)}$, at which point the optimal vertex can be recovered by rounding. The convergence *rate* can be improved: Nesterov and Nemirovskii's *universal barrier* halves the gap every $O(\sqrt{d})$ iterations, where d is the dimension of the feasible region ($d = n - m$ in standard form) [NN94]. The universal barrier is not easy to make algorithmic — evaluating it amounts to computing volumes of polytopes — but significant progress has been made on efficiently computable barriers with nearly the same guarantee, notably the Lewis weight barrier of Lee and Sidford, which achieves $\tilde{O}(\sqrt{d})$ [LS14]. A different kind of guarantee bounds the number of iterations to follow the *entire* central path and solve the LP exactly: Vavasis and Ye's layered least squares method does so in $O(n^{3.5} \log(\tilde{\chi}_A n))$ iterations, where the condition measure $\tilde{\chi}_A$ is essentially our circuit imbalance κ_A [VY96, DHNV20] — a bound independent of b and c altogether.

Two phenomena are not explained by these results. First, in practice interior point methods behave far better than $O(\sqrt{n} \log(1/\epsilon))$: solvers routinely converge in 20–50 iterations essentially regardless of the dimension. Second, on the negative side, Allamigeon, Benchimol, Gaubert and Joswig [ABGJ18] proved that *no* interior point method following the central path is strongly polynomial: there are instances with $O(n)$ constraints on which every path-following method needs $2^{\Omega(n)}$ iterations. How can a method whose iteration count is provably $O(\sqrt{n}L)$ require exponentially many iterations, and what distinguishes the easy instances from the hard ones?

This lecture resolves the tension by changing the question. Instead of asking how many iterations a *specific* method needs, we ask: fixing only the paradigm — *follow the central path* — how many iterations does the *geometry of the path itself* allow? The answer is a concrete quantity, the straight line complexity of the linear program. It is a property of the *instance*, not of any particular method or barrier: it constrains path following with respect to any reasonable notion of central path at once. It can be exponential (Section 4) — proving the lower bound above — but,

as we will show, it is always bounded by a polynomial in n, m and $\log \kappa_A$ (Section 5), which will explain, in Lecture 5, the strongly polynomial solvability of LPs with bounded κ_A encountered in the Lecture 2 landscape.

Throughout this lecture we work with the standard-form pair of Lecture 3,

$$\min c^\top x \text{ s.t. } Ax = b, x \geq \mathbf{0}, \quad \max b^\top y \text{ s.t. } A^\top y + s = c, s \geq \mathbf{0}, \quad (1)$$

with $A \in \mathbb{R}^{m \times n}$ of full row rank, and assume both programs are *strictly feasible* (some $x > \mathbf{0}$ and some $s > \mathbf{0}$ exist). We write v^* for the common optimal value, (x^*, s^*, y^*) for a pair of optimal solutions, and recall from Lecture 1 the gap formula: for feasible x and (s, y) ,

$$c^\top x - b^\top y = (A^\top y + s)^\top x - (Ax)^\top y = s^\top x \geq 0. \quad (2)$$

In particular $s^{*\top} x = c^\top x - v^*$ is the optimality gap of x : the *objective* may be replaced by the nonnegative linear functional $s^{*\top} x$, which vanishes exactly on the optimal face. We write the gap in this form from now on. Strict feasibility of the dual implies that the gap-truncated regions

$$\mathcal{P}_g := \{x : Ax = b, x \geq \mathbf{0}, s^{*\top} x \leq g\}, \quad g \geq 0,$$

are bounded (a routine verification: a strictly feasible slack $s > \mathbf{0}$ bounds $\|x\|_1$ on $\mathcal{P}_g$ via $s^\top x \leq g + s^\top x^*$), which we assume from now on.

2 The Central Path

Interior point methods do not walk on vertices; they traverse the strict interior of the feasible region along a canonical trajectory. For $\mu > 0$, define the *central path point*

$$x^{\text{cp}}(\mu) := \operatorname{argmin} \left\{ c^\top x - \mu \sum_{i=1}^n \ln x_i : Ax = b, x > \mathbf{0} \right\}. \quad (3)$$

The *logarithmic barrier* term blows up at the boundary of the orthant, pushing the minimizer into the interior; the minimizer exists and is unique for every $\mu > 0$ by strict convexity of the objective, whose sublevel sets are compact subsets of the open orthant by our boundedness assumption. For large μ the barrier dominates and $x^{\text{cp}}(\mu)$ sits near the “center” of the feasible region; as $\mu \rightarrow 0$ the objective takes over and $x^{\text{cp}}(\mu)$ converges to an optimal solution x^* . The curve $\mu \mapsto x^{\text{cp}}(\mu)$ is the (*primal*) *central path*; a path-following interior point method starts near $x^{\text{cp}}(\mu_0)$ and tracks the path as $\mu \rightarrow 0$. (Lecture 5 constructs such a method in detail; today only the path itself matters.)

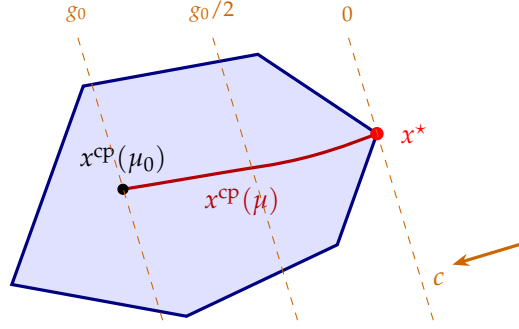


Figure 1: The central path, in the setting of Lecture 2's picture of an approximate solver: it runs from the "center" of the feasible region to the optimal face, staying away from the boundary until the objective forces it in. We are minimizing, so the path moves *against* c . Dashed lines are level sets of the optimality gap $s^{*\top}x$: gap g_0 at the start $x^{\text{cp}}(\mu_0)$, gap 0 at x^* .

The single property of the central path that this lecture uses is that its points are, up to a factor n , as far from the boundary of the orthant as the objective allows.

LEMMA 1 (THE CENTRAL PATH NEARLY MAXIMIZES EVERY COORDINATE) *Let $\mu > 0$ and let $x' \in \mathcal{P}$ satisfy $c^\top x' \leq c^\top x^{\text{cp}}(\mu)$. Then $x'_i \leq n x^{\text{cp}}_i(\mu)$ for every $i \in [n]$.*

PROOF: Write $x := x^{\text{cp}}(\mu)$, an interior minimizer of the convex function $f(w) := c^\top w - \mu \sum_i \ln w_i$ over the affine subspace $\{w : \mathbf{A}w = b\}$. First-order optimality at an interior minimizer gives $\nabla f(x) \perp \ker(\mathbf{A})$, and since $x' - x \in \ker(\mathbf{A})$,

$$0 = \left(c - \mu x^{-1}\right)^\top (x' - x) = c^\top x' - c^\top x - \mu \sum_{i=1}^n \frac{x'_i}{x_i} + \mu n,$$

where $x^{-1} := (1/x_1, \dots, 1/x_n)$. Using $c^\top x' \leq c^\top x$, we conclude $\sum_{i=1}^n x'_i/x_i \leq n$. As every term is nonnegative, each single term is at most n . $\square$

Path-following methods, abstractly. Whatever their internal details, path-following interior point methods share one behavior: they generate a sequence of interior iterates that remain *multiplicatively* close to the central path — at each iterate, the slack of every inequality is within a factor γ of the slack achieved by the central path point with the same optimality gap — and consecutive iterates are joined by segments that also stay in this multiplicative tube. In standard form the slacks are the coordinates x_i themselves; the same statement makes sense verbatim in geometric form, where the slack of $a_i^\top x \leq b_i$ is $b_i - a_i^\top x$. (Lecture 5's method stays in such a tube with γ a constant; wider tubes only help the lower bounds below.) The trajectory of a T -iteration run is thus a piecewise linear curve with T pieces inside the tube around the path. Figure 2 shows the tube and such a curve on a concrete instance. Lower-bounding T therefore reduces to a purely geometric question: *how few pieces can a piecewise linear curve in the tube have?* To answer it, we now replace the central path — an analytic object, defined by an optimization problem we cannot easily solve — by a combinatorial proxy.

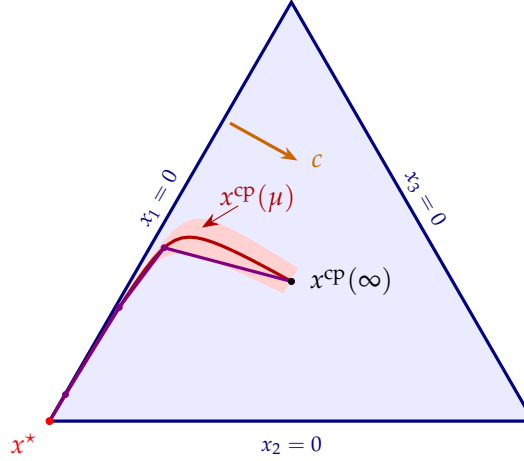


Figure 2: Path following in a tube, computed exactly for the LP $\min x_1 + \varepsilon x_2 + \varepsilon^2 x_3$ over the simplex $x_1 + x_2 + x_3 = 1$, $x \geq \mathbf{0}$, with $\varepsilon = 1/50$, drawn via the isometric embedding of the plane $\{x \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 1\}$ into $\mathbb{R}^2$. The feasible region becomes an equilateral triangle whose edges are the facets $x_i = 0$, and each slack x_i is proportional to the distance to the corresponding edge, with the same constant for all three. The red curve is the central path, running from the analytic center $x^{\text{CP}}(\infty) = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ — the centroid of the triangle — to the optimal vertex $x^* = e_3$; the shaded region is the multiplicative tube of factor 1.1: all points whose three slacks are within a factor 1.1 of the slacks of the central path point with the same optimality gap. The violet curve is a piecewise linear path from $x^{\text{CP}}(\infty)$ to x^* staying inside the tube, with $T = 4$ segments (chosen greedily); its last segment reaches x^* along the asymptotic direction of the path. The tube is wide where the path is straight and pinches at the elbows — near the elbows is where segments are spent.

3 The Max Central Path and Straight Line Complexity

DEFINITION 2 (MAX CENTRAL PATH) For $i \in [n]$ and $g \geq 0$, the max central path (in coordinate i) is

$$x_i^{\text{m}}(g) := \max\{x_i : x \in \mathcal{P}_g\} = \max\{x_i : \mathbf{A}x = b, x \geq \mathbf{0}, s^{*\top}x \leq g\},$$

the largest i -th coordinate compatible with optimality gap g .

Each x_i^{m} is a nondecreasing, concave (interpolating maximizers for two gap values produces a feasible point for the intermediate gap), piecewise linear function, with $x_i^{\text{m}}(0)$ the largest i -th coordinate of an optimal solution. Note that the vector $x^{\text{m}}(g) = (x_1^{\text{m}}(g), \dots, x_n^{\text{m}}(g))$ is generally *not* feasible — each coordinate is the value of a different program — but this will not matter: the max central path is a bookkeeping device, one concave curve per coordinate. Its first virtue is that it pins down the central path up to a factor n . Write $g_\mu := s^{*\top}x^{\text{CP}}(\mu)$ for the gap of the central path point at parameter μ (an increasing function of μ).

COROLLARY 3 (SANDWICH) For every $\mu > 0$ and $i \in [n]$:

$$\frac{1}{n} x_i^{\text{m}}(g_\mu) \leq x_i^{\text{CP}}(\mu) \leq x_i^{\text{m}}(g_\mu).$$

PROOF: The upper bound holds as $x^{\text{CP}}(\mu) \in \mathcal{P}_{g_\mu}$. For the lower bound, the maximizer x' attaining $x_i^{\text{m}}(g_\mu)$ satisfies $s^{*\top} x' \leq g_\mu = s^{*\top} x^{\text{CP}}(\mu)$, equivalently $c^\top x' \leq c^\top x^{\text{CP}}(\mu)$, so Lemma 1 gives $x_i^{\text{m}}(g_\mu) = x'_i \leq n x_i^{\text{CP}}(\mu)$. $\square$

Thus the central path threads the multiplicative tube of width n around the max central path, coordinate by coordinate. The quantity that measures how “straight” a curve is within multiplicative error is the following.

DEFINITION 4 (STRAIGHT LINE COMPLEXITY) Let $f : [0, g_0] \rightarrow \mathbb{R}_+$ and $\gamma \in (0, 1]$. The straight line complexity $\text{SLC}_\gamma(f, g_0)$ is the minimum number of pieces of a continuous piecewise linear function $h : [0, g_0] \rightarrow \mathbb{R}_+$ satisfying $\gamma f \leq h \leq f$ on $[0, g_0]$.

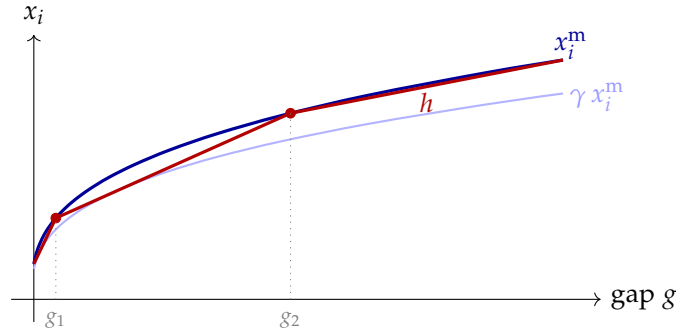


Figure 3: Straight line complexity: the minimum number of linear pieces of a curve h squeezed between γx_i^{m} and x_i^{m} . Here $x_i^{\text{m}}(g) = 1.6(g + 0.03)^{0.35}$ and $\gamma = 0.86$, and h is built by the greedy algorithm of Exercise 2: each chord of x_i^{m} extends maximally before it would dip below γx_i^{m} , producing breakpoints $g_1 \approx 0.29$ and $g_2 \approx 3.40$. The greedy construction is optimal, so $\text{SLC}_\gamma(x_i^{\text{m}}, g_0) = 3$ exactly.

The definition is stable in exactly the ways we need: reparametrizing the argument (e.g., trading the gap g for the path parameter $\mu = \Theta(g/n)$) does not change it, and scaling the approximation factor from γ to γ/n costs nothing qualitatively — the tube just widens. The promised lower bound is now immediate.

PROPOSITION 5 (SLC LOWER-BOUNDS PATH FOLLOWING) Let $\gamma \in (0, 1]$, and let $x(t)$, $t \in [0, 1]$, be a piecewise linear curve in $\mathcal{P}$ with T pieces, with strictly decreasing gap, running from gap g_0 to 0, whose points satisfy the tube condition $\gamma x_i^{\text{CP}}(\mu(t)) \leq x_i(t) \leq \gamma^{-1} x_i^{\text{CP}}(\mu(t))$ for all i , where $\mu(t)$ is the path parameter with matching gap. Then

$$T \geq \max_{i \in [n]} \text{SLC}_{\gamma^2/n}(x_i^{\text{m}}, g_0).$$

PROOF: Fix i . Along each of the T segments, the gap $g(t) = s^{*\top} x(t)$ is an affine function of t , and strictly decreasing; hence coordinate i of the trajectory, expressed as a function of the gap, is a piecewise linear function $\tilde{h}(g)$ with at most T pieces. By the tube condition and Corollary 3,

$$\frac{\gamma}{n} x_i^{\text{m}}(g) \leq \gamma x_i^{\text{CP}} \leq \tilde{h}(g) \leq \gamma^{-1} x_i^{\text{CP}} \leq \gamma^{-1} x_i^{\text{m}}(g).$$

Then $h := \gamma \tilde{h}$ satisfies $(\gamma^2/n) x_i^{\text{m}} \leq h \leq x_i^{\text{m}}$, so $T \geq \text{SLC}_{\gamma^2/n}(x_i^{\text{m}}, g_0)$. $\square$

3.1 The max central path is a shadow

The max central path has a familiar geometric meaning. Project the feasible region onto the plane spanned by the gap and coordinate i :

$$\mathcal{P}^i := \{(s^{*\top}x, x_i) : x \in \mathcal{P}\} \subseteq \mathbb{R}_+^2.$$

This is a two-dimensional shadow of the type studied in Lectures 2 and 3, and by definition $x_i^m(g) = \max\{y : (t, y) \in \mathcal{P}^i, t \leq g\}$: the graph of x_i^m is precisely the increasing part of the *upper boundary* of the shadow $\mathcal{P}^i$ (followed by a constant piece once the maximum of x_i over $\mathcal{P}$ is reached). In particular, exactly as in Lecture 2, the breakpoints of x_i^m are projections of vertices of $\mathcal{P}$, joined by projections of edges: the max central path traces a *shadow vertex simplex path*. Two consequences:

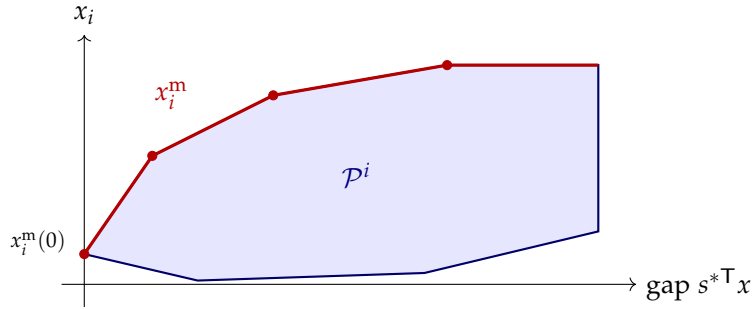


Figure 4: The max central path x_i^m is the increasing upper boundary of the two-dimensional shadow $\mathcal{P}^i$ — the graph of a shadow vertex simplex path, with breakpoints at projected vertices.

First, an unconditional — if enormous — upper bound: the number of pieces of x_i^m is at most the number of vertices of $\mathcal{P}$ plus one, so

$$\text{SLC}_1(x_i^m, g_0) \leq \binom{n}{m} + 1 \quad \text{for every } i, g_0.$$

Second, a conceptual bridge: the simplex-path geometry that governed the smoothed analyses of Lecture 3 turns out to also govern the iteration complexity of interior point methods. This bridge carries traffic in both directions, by the following theorem — proved via the algorithm of Lecture 5 — which complements Proposition 5 with a nearly matching *upper* bound.

THEOREM 6 (IPMS MATCH THE GEOMETRY OF THE PATH [ADL⁺22]) *There is a path-following interior point method (the TRUST-REGION IPM of Lecture 5) which, started at a point close to the central path at parameter μ_0 (of gap $g_0 = n\mu_0$; Lecture 5 makes this precise), computes an exact optimal primal-dual pair within*

$$O\left(\min_{\gamma \in (0,1]} \sqrt{n} \log\left(\frac{n}{\gamma}\right) \sum_{i=1}^n \text{SLC}_\gamma(x_i^m, g_0)\right)$$

iterations, each iteration solving $O(1)$ linear systems.

Combined with Proposition 5, the straight line complexity of the max central path thus determines the iteration complexity of path-following, up to a factor $\tilde{O}(n^{1.5})$: it is both the floor for every method and, within this factor, the ceiling for the best one.

Theorem 6 is not only a conceptual statement; it is an engine for algorithmic results. The main theorem of [DKN⁺24] builds on it: for any primal-dual pair of the form (1) in which $\mathbf{A}$ has at most two nonzero entries per column, there is a *strongly polynomial* algorithm that computes optimal primal and dual solutions, or a certificate of infeasibility of one of the two programs. The route passes through bounding the straight line complexity of such instances; Section 5 presents the core of this technique.

The rest of the lecture computes this quantity: we show it can be exponential (Section 4), and that circuits bound it (Section 5).

Exercise 1 (The max central path of maximum flow) Let $\mathcal{P} \subseteq \mathbb{R}_+^E$ be the feasible region of the maximum flow problem of Lecture 3 on a network $G = (V, E)$ with source s , sink t and capacities u , with flow value $\text{val}(f)$ and optimal value v^* . For $e \in E$, the max central path on e is

$$f_e^{\text{m}}(g) := \max\{f_e : f \in \mathcal{P}, \text{val}(f) \geq v^* - g\}.$$

Prove that f_e^{m} is piecewise linear with at most 2 pieces, i.e., $\text{SLC}_1(f_e^{\text{m}}) \leq 2$. (Hint: Adjacent vertices f^1, f^2 of $\mathcal{P}$ with $\text{val}(f^1) > \text{val}(f^2)$ differ by a t - s path in the residual network $N[f^1]$ of Lecture 3.)

4 Straight Line Complexity Can Be Exponential

We now prove — in outline, following [ABGJ18] — that the straight line complexity of the max central path can be exponential in the dimension, even for instances with $O(n)$ variables and constraints. By Proposition 5, this shows that *every* path-following interior point method has exponential worst-case iteration complexity, and hence that no such method is strongly polynomial.

The tropical program. The construction starts not from a linear program but from a *tropical linear program*. Consider the following optimization problem in the variables $y_1, \dots, y_{2n+1}$:

$$\begin{aligned} \min \quad & y_1 \\ \text{s.t.} \quad & y_1 \leq 2, \quad y_2 \leq 1, \\ & y_{2j+1} \leq 1 + \min\{y_{2j-1}, y_{2j}\}, \quad 1 \leq j \leq n, \\ & y_{2j+2} \leq 1 - \frac{1}{2j} + \max\{y_{2j-1}, y_{2j}\}, \quad 1 \leq j < n. \end{aligned} \tag{4}$$

Programs built from affine forms, min and max are called *tropical*: they are linear over the semiring in which “addition” is max and “multiplication” is $+$. Note where the program fails to be linear in the ordinary sense. A min-constraint splits into two separate linear inequalities — $y_{2j+1} \leq 1 + \min\{y_{2j-1}, y_{2j}\}$ is exactly the pair $y_{2j+1} \leq 1 + y_{2j-1}$ and $y_{2j+1} \leq 1 + y_{2j}$ — so those constraints are linear. The max-constraints $y_{2j+2} \leq 1 - \frac{1}{2j} + \max\{y_{2j-1}, y_{2j}\}$ are not: they require y_{2j+2} to be below at least *one* of two linear functions, a union of two half-spaces, which is non-convex and cannot be expressed by any finite system of linear inequalities. In exact analogy with Definition 2, define the *tropical max central path* of (4) by

$$y_j^{\text{m}}(g) := \max\{y_j : y \text{ feasible for (4), } y_1 \leq g\}, \quad g \in [0, 2].$$

Because every constraint bounds a variable from above by a nondecreasing expression in *earlier* variables, the maxima decouple and the recursion unrolls exactly: $y_1^m(g) = g$, $y_2^m = 1$, and

$$y_{2j+1}^m = 1 + \min\{y_{2j-1}^m, y_{2j}^m\}, \quad y_{2j+2}^m = 1 - \frac{1}{2^j} + \max\{y_{2j-1}^m, y_{2j}^m\}.$$

Everything about this recursion can be computed in closed form.

LEMMA 7 (THE TROPICAL STAIRCASE) *On $g \in [0, 2]$, every y_j^m is nondecreasing and piecewise linear with all slopes in $\{0, 1\}$. For each level $j \geq 1$, the curves y_{2j+1}^m and y_{2j+2}^m (the latter for $j < n$) have exactly 2^j linear pieces, with breakpoints at the multiples of 2^{1-j} , alternating slope 1 and slope 0; the odd coordinate starts with a rise, the even one with a flat. In particular, the top coordinate y_{2n+1}^m is a staircase with exactly 2^n pieces: 2^{n-1} rises of height 2^{1-n} , alternating with 2^{n-1} flats.*

The doubling mechanism is the combinatorial heart of the construction: the max-constraint lets y_{2j+2} inherit the better of its two predecessors, and its geometrically shrinking offset $1/2^j$ alternates *which* predecessor is better as g varies; the min-constraint interleaves the same two curves the other way. Each level thus doubles the breakpoints of the previous one (Figure 5). We leave the verification of the first levels to the reader, and refer to [ABGJ18] for the general induction.

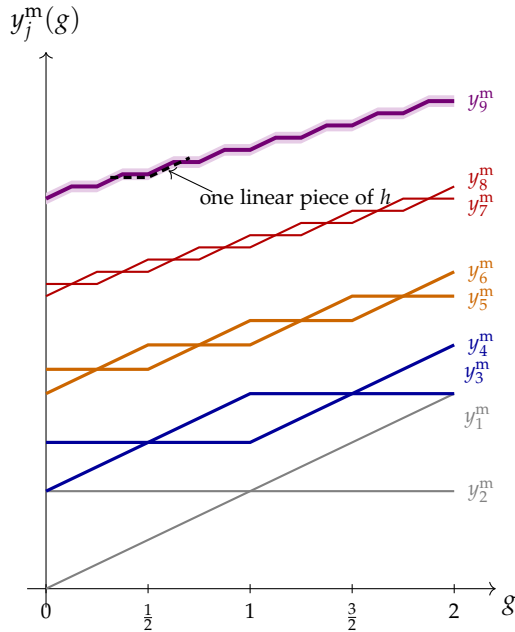


Figure 5: The tropical max central path of (4) for $n = 4$, computed exactly from the recursion. Each level doubles the pieces of the previous one — 2, 4, 8 for the blue, orange and red pairs — and the top coordinate $y_{2n+1}^m = y_9^m$ (violet) is a staircase with $2^n = 16$ pieces, per Lemma 7. By Step 1 below, the LP max central path coordinate $\log_t x_{2n+1}^m(t^g)$ is glued to this staircase within the narrow shaded band. On this scale a linear piece of an approximating curve h is a convex hockey stick $\max\{g + \log_t a, \log_t b\}$ (dashed): it can cover one flat and the following rise, but never passes a concave corner, so tracking the whole staircase needs $2^{n-1} - 1 = 7$ pieces (Step 2).

From tropical to linear. The tropical program lifts to a linear program: replace each variable y_j by $x_j := t^{y_j}$, where $t > 1$ is a large base, so that additive constants become factors ($1 + y \mapsto t x$) and each max becomes — approximately — a sum. The approximation is governed by the *softmax inequalities*

$$\max\{a, b\} \leq \log_t(t^a + t^b) \leq \max\{a, b\} + \log_t 2, \quad (5)$$

which follow from $t^{\max\{a,b\}} \leq t^a + t^b \leq 2 t^{\max\{a,b\}}$: on the $\log_t$ scale, a sum is a max, up to an additive error $\log_t 2$ that vanishes as t grows. (The min-constraints need no approximation at all: $y_{2j+1} \leq 1 + \min\{y_{2j-1}, y_{2j}\}$ is simply the pair of linear constraints $x_{2j+1} \leq t x_{2j-1}$ and $x_{2j+1} \leq t x_{2j}$.) The resulting linear program appears in the following theorem, which asserts that its max central path inherits the staircase geometry of Lemma 7.

THEOREM 8 ([ABGJ18]) *For a parameter $t > 1$, consider the linear program $\text{LP}(t)$ in the variables $x_1, \dots, x_{2n+1}$:*

$$\begin{aligned} \min \quad & x_1 \\ \text{s.t.} \quad & x_1 \leq t^2, \quad x_2 \leq t, \\ & x_{2j+1} \leq t x_{2j-1}, \quad x_{2j+1} \leq t x_{2j}, \quad 1 \leq j \leq n, \\ & x_{2j+2} \leq t^{1-1/2^j} (x_{2j-1} + x_{2j}), \quad 1 \leq j < n, \\ & x_{2n-1} \geq 0, \quad x_{2n} \geq 0, \quad x_{2n+1} \geq 0, \end{aligned}$$

whose optimal value is 0, so that the optimality gap of a feasible x is x_1 . (In fact all variables are implicitly nonnegative: $0 \leq x_{2j+1} \leq t x_{2j-1}$ and $x_{2j+1} \leq t x_{2j}$ propagate the sign constraint on x_{2n-1} down to every lower-indexed variable; only the top variable x_{2n+1} needs its own.) For every $\gamma \in (0, 1)$ and $t \geq 2^{\Omega(2^n(n+\log(1/\gamma)))}$, the max central path coordinate of $\text{LP}(t)$ satisfies $\text{SLC}_\gamma(x_{2n+1}^m) \geq 2^{n-1} - 1$.

Note the shape of the statement: the instance has $O(n)$ constraints, but the parameter t — hence the bit size $L \approx \log t$ — is doubly exponential in n . This is consistent with the weakly polynomial bound $O(\sqrt{n}L)$: exponentially many iterations require exponential bit size. The theorem thus separates strongly polynomial behavior from weakly polynomial behavior exactly. The proof has two steps: for very large t , the max central path of $\text{LP}(t)$ collapses, on the $\log_t$ scale, onto the tropical staircase; and no piecewise linear curve with few pieces can track a fine staircase.

Step 1: the collapse. The softmax errors accumulate along the recursion, but only linearly: comparing the max central path of $\text{LP}(t)$ with the tropical one, an induction through the layers of constraints gives

$$\left| \log_t x_j^m(t^g) - y_j^m(g) \right| \leq j \log_t 2, \quad g \in [0, 2]. \quad (6)$$

For t doubly exponential in n , the total error $(2n+1) \log_t 2$ is far smaller than 2^{-n} — the LP max central path, on the $\log_t$ scale, is glued to the staircase to within a small fraction of a single step.

Step 2: staircases defeat lines. Let h be piecewise linear with T pieces and $\gamma x_{2n+1}^m \leq h \leq x_{2n+1}^m$ on the relevant gap range. Passing to the $\log_t$ scale, $H(g) := \log_t h(t^g)$ lies, by (6), in an additive tube of half-width $\delta := \log_t(1/\gamma) + (2n+1) \log_t 2$ around the staircase y_{2n+1}^m , and $\delta \ll 2^{-n}$ for t as in the theorem.

Consider a single linear piece $h(\xi) = a\xi + b$ of h , where $\xi = t^g$ is the gap on the original scale. We may assume $a, b \geq 0$ for every piece. Indeed, since x_{2n+1}^m is nondecreasing, replacing h by its

running maximum $\xi \mapsto \max_{\xi' \leq \xi} h(\xi')$ keeps it inside the tube without increasing the number of pieces, so all slopes a may be taken nonnegative; and a piece with $b < 0$ has, on the $\log_t$ scale, derivative $H'(g) = at^g / (at^g + b) > 1$ everywhere — a curve that is never flat and so can track at most a single rise of the staircase, which is weaker than what we are about to allow the pieces with $b \geq 0$. For $a, b \geq 0$, the softmax inequality (5) gives

$$H(g) = \log_t (a t^g + b) \in \max\{g + \log_t a, \log_t b\} + [0, \log_t 2] :$$

up to negligible error, a *flat piece followed by a slope-one piece* — a convex “hockey stick” with a single upward corner. Inside a tube of half-width $\delta \ll$ step size, a hockey stick can cover at most one flat of the staircase followed by one rise; it can never pass a *concave* (rise-to-flat) corner, since its slope never decreases. By Lemma 7, the staircase y_{2n+1}^m contains $2^{n-1} - 1$ disjoint flat-rise pairs in the interior of $[0, 2]$, and each needs its own piece of h : hence $T \geq 2^{n-1} - 1$. $\square$ (outline)

Combining Theorem 8 with Proposition 5 and Theorem 6: no path-following interior point method terminates in fewer than $2^{n-1} - 1$ iterations on $\text{LP}(t)$, while the trust-region method of Lecture 5 needs at most $\tilde{O}(\sqrt{n} 2^n)$ — both floor and ceiling are exponential here, as promised by the tightness of the SLC characterization.

Exercise 2 (Greedy and staircase approximation) Let $f : [0, T] \rightarrow \mathbb{R}_+$ be concave and nondecreasing and let $\gamma \in (0, 1]$. For $0 \leq a < b \leq T$, let $\ell_{a,b}^f$ denote the linear interpolation of f on $[a, b]$, and set $s_b^f(g) := f(b) \min\{g/b, 1\}$ and $s_0^f \equiv f(0)$.

1. Prove that the following greedy construction is optimal, achieving $\text{SLC}_\gamma(f, T)$ pieces: build breakpoints $0 = g_0 < g_1 < \dots < g_k = T$, taking $g_i \in (g_{i-1}, T]$ maximal with $\gamma f \leq \ell_{g_{i-1}, g_i}^f \leq f$ on $[g_{i-1}, g_i]$, and let $h := \ell_{g_{i-1}, g_i}^f$ there. (Hint: For any valid p with breakpoints $0 = b_0 < \dots < b_r = T$, show by induction that $i \leq r$ and $b_i \leq g_i$.)
2. Let $h^s := \max_{0 \leq i \leq k} s_{g_i}^f$. Prove that $\frac{1}{2} h \leq h^s \leq h$ on $[0, T]$: up to a factor 2 in the number of pieces and in the approximation factor, $\text{SLC}_\gamma(f, T)$ is achieved by a staircase.

Exercise 3 (Duality of the max central path) Define the dual max central path by

$$s_i^m(g) := \max\{s_i : (s, y) \text{ dual feasible, } b^\top y \geq v^* - g\}.$$

1. Prove that $g \leq x_i^m(g) s_i^m(g) \leq 2g$ for all $g \geq 0$ — a combinatorial analogue of the centrality equation $x_i s_i = \mu$ of Lecture 5. (Hint: Upper bound: the gap formula (2). Lower bound: dualize both maximization problems and multiply the optimal values.)
2. Combine part 1 with the staircases of Exercise 2 to convince yourself — no formal proof needed — that x_i^m and s_i^m have the same straight line complexity up to constant factors in both parameters. (Hint: What does $g \mapsto g/s_b^f(g)$ look like?)

5 Bounding SLC via Circuits

The lower bound instance was carefully cooked; we now show what makes it exceptional. The straight line complexity of the max central path is controlled by the *circuits* of $\mathbf{A}$ (Lecture 3): whenever the circuit imbalance $\kappa = \kappa_{\mathbf{A}}$ is moderate, the SLC is at most $\min\{m, n - m\} + 2$ — and hence path-following IPMs need only polynomially many iterations, with *no dependence on b and c whatsoever*. Throughout this section, fix $i \in [n]$; we bound $\text{SLC}(x_i^m)$ over all of $\mathbb{R}_+$, uniformly in g_0 .

Preprocessing. Fix a *basic* (i.e., vertex) optimal primal solution x^* attaining $x_i^* = x_i^m(0)$, and any basic optimal dual solution (s^*, y^*) ; thus $|\text{supp}(x^*)| \leq m$ and $|\text{supp}(s^*)| \leq n - m$, and by complementary slackness the supports are disjoint. By the gap formula (2), membership in $\mathcal{P}_g$ reads $s^{*\top} x \leq g$, so only the vector $s^* \geq \mathbf{0}$ enters the definition of x_i^m .

5.1 Circuit curves

The elementary moves available from x^* are the circuits of $\mathbf{A}$. For $h \in \ker(\mathbf{A})$, define the h -curve

$$\bar{x}^h(g) := x^* + \alpha^h(g) h, \quad \alpha^h(g) := \max\{\alpha \geq 0 : x^* + \alpha h \geq \mathbf{0}, \alpha s^{*\top} h \leq g\},$$

the farthest point of $\mathcal{P}_g$ reachable from x^* along direction h . (If $x^* + \alpha h \geq \mathbf{0}$ for some $\alpha > 0$, then optimality of x^* forces $s^{*\top} h = c^\top h \geq 0$, so the curve is well defined; boundedness of $\mathcal{P}_g$ rules out $\alpha^h(g) = \infty$.) Explicitly,

$$\alpha^h(g) = \min \left(\underbrace{\frac{g}{s^{*\top} h}}_{\text{gap budget}}, \underbrace{\min_{j: h_j < 0} \frac{x_j^*}{|h_j|}}_{\text{first blocking coordinate}} \right), \quad (7)$$

with the conventions that the first term is $+\infty$ if $s^{*\top} h = 0$ and the second if $h \geq \mathbf{0}$. As a function of g , the coordinate $\bar{x}_i^h(g)$ is thus a concave *two-piece* curve: it rises linearly from x_i^* with slope $h_i/(s^{*\top} h)$ until the moving point hits the boundary of the orthant, and stays constant thereafter (Figure 6).

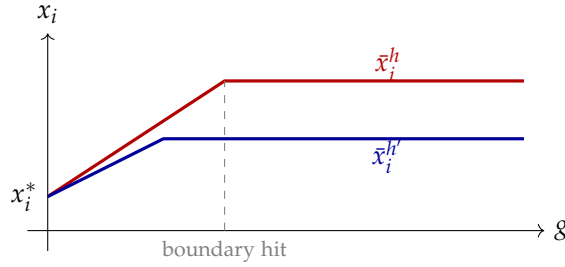


Figure 6: Two circuit curves in coordinate i : each rises at slope $h_i/(s^{*\top} h)$ from x_i^* and flattens when a coordinate of $x^* + \alpha h$ reaches 0.

Circuit curves capture the max central path up to a factor n — the proof is our third application of conformal circuit decomposition (Lecture 3), after flow decomposition and the anchoring of shadow paths.

LEMMA 9 (CIRCUIT CURVES CAPTURE THE MAX CENTRAL PATH) *For every $g \geq 0$ with $x_i^m(g) > x_i^*$, there exists a scaled circuit $h \in \ker(\mathbf{A})$ with $h_i > 0$, $h_j \geq 0$ whenever $x_j^* = 0$, and*

$$\bar{x}_i^h(g) \geq \frac{x_i^m(g)}{n}.$$

PROOF: Let $\hat{x} \in \mathcal{P}_g$ attain $\hat{x}_i = x_i^m(g)$, and take a conformal circuit decomposition $\hat{x} - x^* = \sum_{j=1}^{\ell} h^{(j)}$ with $\ell \leq n$ (Lecture 3; the bound $\ell \leq n$ follows by keeping only linearly independent terms, via Carathéodory's theorem of Lecture 1). By conformality, each partial step stays in the

box between x^* and $\hat{x}$: in particular $x^* + h^{(j)} \geq \mathbf{0}$ is feasible, so $c^\top h^{(j)} \geq 0$ by optimality of x^* , and $s^{*\top} h^{(j)} \leq s^{*\top}(\hat{x} - x^*) \leq g$, so $x^* + h^{(j)} \in \mathcal{P}_g$ and $\alpha^{h^{(j)}}(g) \geq 1$. Choosing k maximizing $h_i^{(k)}$,

$$\bar{x}_i^{h^{(k)}}(g) \geq x_i^* + h_i^{(k)} \geq x_i^* + \frac{1}{\ell} \sum_{j=1}^{\ell} h_i^{(j)} \geq \frac{\hat{x}_i}{\ell} \geq \frac{x_i^m(g)}{n}.$$

Finally $h_i^{(k)} > 0$ since $\hat{x}_i > x_i^*$, and $h_j^{(k)} \geq 0$ whenever $x_j^* = 0$ since $x^* + h^{(k)} \geq \mathbf{0}$. $\square$

5.2 Circuit covers

By Lemma 9, the pointwise maximum of *all* circuit curves $1/n$ -approximates x_i^m — but there are exponentially many circuits. The key definition asks for a small subfamily whose curves dominate all others.

DEFINITION 10 (CIRCUIT COVER) *A finite set $S \subseteq \ker(\mathbf{A})$ with $c^\top h \geq 0$ for all $h \in S$ is a γ -circuit cover of i (with respect to x^*) if for every scaled circuit h' with $c^\top h' \geq 0$ there exists $h \in S$ with $\bar{x}_i^h \geq \gamma \bar{x}_i^{h'}$ pointwise.*

LEMMA 11 (COVERS BOUND SLC) *If S is a γ -circuit cover of i with respect to x^* (with $x_i^* = x_i^m(0)$), then*

$$\text{SLC}_{\gamma/n}(x_i^m) \leq |S| + 1.$$

PROOF: Let $\bar{x}_i^S(g) := \max_{h \in S} \bar{x}_i^h(g)$, and let $h^\uparrow$ be its *upper concave envelope* (Figure 7). Since each $\bar{x}_i^h$ is concave with two pieces, the envelope of $|S|$ such curves is concave piecewise linear with at most $|S| + 1$ pieces: sorted by their initial slopes, each curve can contribute at most one linear piece to the envelope, plus one final constant piece. By Lemma 9 and the cover property, $\bar{x}_i^S \geq (\gamma/n) x_i^m$ pointwise, while trivially $\bar{x}_i^S \leq x_i^m$; since x_i^m is concave, the envelope $h^\uparrow$ also satisfies $(\gamma/n) x_i^m \leq h^\uparrow \leq x_i^m$. $\square$

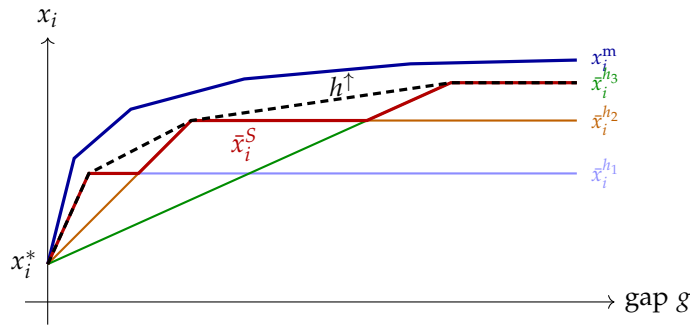


Figure 7: The curves of a circuit cover $S = \{h_1, h_2, h_3\}$ (thin), their pointwise maximum $\bar{x}_i^S$ (thick red), and its upper concave envelope $h^\uparrow$ (dashed), which bridges the non-concave dents of the maximum. Sorted by initial slope, each cover curve contributes at most one rising piece to the envelope, plus one final flat: $h^\uparrow$ is concave piecewise linear with at most $|S| + 1$ pieces (here 4). All curves start at $x_i^* = x_i^m(0)$, and $(\gamma/n) x_i^m \leq \bar{x}_i^S \leq h^\uparrow \leq x_i^m$: the maximum of the cover curves tracks the max central path x_i^m (blue) up to the factor γ/n .

For which matrices do the circuits have enough combinatorial structure to build small covers from? The prime example is *generalized flow*, whose circuits the next exercise characterizes. Building a circuit cover for exactly this family of circuits is the main combinatorial work of the strongly polynomial algorithm of [DKN⁺24] for LPs whose constraint matrix has at most two nonzero entries per column: a simple reduction, at the cost of doubling the number of variables, transforms any such LP into one whose constraint matrix is a generalized incidence matrix as below. The full characterization of these circuits, for arbitrary gains over any field, is due to Zaslavsky [Zas91].

Concretely, let $G = (V, E)$ be a directed graph whose edges $e = (i, j)$ carry gain factors $\gamma_e > 0$. The *generalized incidence matrix* $\mathbf{A} \in \mathbb{R}^{V \times E}$ has, in the column of $e = (i, j)$, the entry -1 in row i and γ_e in row j (for $\gamma \equiv \mathbf{1}$, the node-arc incidence matrix of Lecture 3); a kernel vector is a *generalized circulation* — f_e units entering e deliver $\gamma_e f_e$ at its head. The *gain* of a cycle C with a chosen traversal direction is $\gamma(C) := \prod_{e \text{ forward}} \gamma_e \cdot \prod_{e \text{ backward}} \gamma_e^{-1}$, and C is *conservative* if $\gamma(C) = 1$. Finally, a *bicycle* is a connected subgraph with exactly two more edges than a spanning tree: two cycles meeting at one node, two disjoint cycles joined by a path, or a theta graph. When its cycles are non-conservative, a bicycle carries a natural flow: a *flow-generating* cycle ships its surplus along the path to a *flow-absorbing* cycle.

Exercise 4 (Circuits of generalized flow) Show that the supports of the circuits of $\mathbf{A}$ are exactly (i) the conservative cycles, and (ii) the bicycles containing no conservative cycle. (Hint: (i) Push one unit around a cycle C : one lap multiplies the flow by $\gamma(C)$. (ii) Forests, and connected subgraphs whose only cycle is non-conservative, have independent columns. (iii) On a bicycle, generate a surplus on one cycle and absorb it on the other.)

5.3 A small cover from the circuit imbalance

It remains to construct a small cover — this is where $\kappa = \kappa_{\mathbf{A}}$ and the Pareto-signature argument enter.

THEOREM 12 (SLC VIA CIRCUIT IMBALANCE [DKN⁺24]) *For every $i \in [n]$ there is a $(1/(n\kappa^2))$ -circuit cover of i of size at most $\min\{m, n - m\} + 1$. Consequently,*

$$\text{SLC}_{1/(n^2\kappa^2)}(x_i^m) \leq \min\{m, n - m\} + 2 \quad \text{for every } i \in [n].$$

PROOF: Order the coordinates so that $x_1^* \geq x_2^* \geq \dots \geq x_n^*$ and $s_1^* \leq \dots \leq s_n^*$ (possible by disjointness of supports), and let $p := |\text{supp}(x^*)| \leq m$ and $d := \min\{j : s_j^* > 0\} \geq n - (n - m) + 1$, so that $n - d + 1 \leq n - m$.

Restrict attention to the circuits that matter: by (7) and Lemma 9, if $h_i \leq 0$, or $h_j < 0$ for some j with $x_j^* = 0$, then $\bar{x}_i^h \leq x_i^*$ pointwise (in the second case $\alpha^h \equiv 0$), so the curve is dominated by every other curve; we may thus normalize $h_i = 1$ and assume $h_j \geq 0$ for $j > p$. Call the set of such scaled circuits $\mathcal{C}'$; note that entries of $h \in \mathcal{C}'$ satisfy

$$1/\kappa \leq |h_j| \leq \kappa \quad \text{for all } j \in \text{supp}(h), \tag{8}$$

by the definition of κ and $h_i = 1$. Assign to $h \in \mathcal{C}'$ the *signature*

$$I^p(h) := \max(\{0\} \cup \{j \leq p : h_j < 0\}), \quad I^d(h) := \max\{j \geq d : h_j > 0\} :$$

the deepest (smallest- x^*) coordinate that blocks the step, and the most expensive dual coordinate the step pays for. The set defining I^d is never empty — so $I^d(h) \geq d$ — since otherwise $h_j = 0$ for all $j \geq d$ (recall $h_j \geq 0$ there), giving $s^{*\top}h = 0$: then $x^* + \alpha h$ would be optimal for small $\alpha > 0$ with i -th coordinate exceeding $x_i^* = x_i^m(0)$, contradicting the choice of x^* . Let S contain one representative for each Pareto-minimal signature (no other signature is coordinate-wise smaller). Distinct members of S have distinct values of I^p (else one would dominate the other), and I^p takes at most $p + 1 \leq m + 1$ values; likewise at most $n - d + 1 \leq n - m$ values of I^d ; hence $|S| \leq \min\{m + 1, n - m\} \leq \min\{m, n - m\} + 1$ (Figure 8).

We claim S is a $1/(n\kappa^2)$ -cover. Let $h \in \mathcal{C}'$ and pick $h' \in S$ with $I^p(h') \leq I^p(h)$ and $I^d(h') \leq I^d(h)$. By the two-piece structure (7), it suffices to compare the initial slope $1/(s^{*\top}h)$ and the cap $\min_{j:h_j < 0} x_j^* / |h_j|$ of the two curves.

Slopes. Write $j' := I^d(h') \leq j'' := I^d(h)$; both are at least d , as noted above. Using that h' has at most n positive entries in $[d, n]$, all of size $\leq \kappa$, while h has a positive entry of size $\geq 1/\kappa$ at j'' and only nonnegative entries beyond p :

$$s^{*\top}h' \leq n\kappa s_{j'}^* \leq n\kappa s_{j''}^* \leq n\kappa^2 s_{j''}^* h_{j''} \leq n\kappa^2 s^{*\top}h.$$

So the slope of $\bar{x}_i^{h'}$ is at least $1/(n\kappa^2)$ times that of $\bar{x}_i^h$.

Caps. If $I^p(h') = 0$ then $h' \geq \mathbf{0}$ and the cap of h' is infinite. Otherwise let $\ell' \leq I^p(h') =: i'$ attain the cap of h' and $\ell \leq I^p(h) =: i''$ attain that of h ; note $i' \leq i''$, so $x_{\ell'}^* \geq x_{i'}^* \geq x_{i''}^*$ by the ordering. Then

$$\frac{x_{\ell'}^*}{|h'_{\ell'}|} \geq \frac{x_{i''}^*}{\kappa} \geq \frac{1}{\kappa^2} \cdot \frac{x_{i''}^*}{|h_{i''}|} \geq \frac{1}{\kappa^2} \cdot \min_{j:h_j < 0} \frac{x_j^*}{|h_j|},$$

using (8) twice. So the cap of $\bar{x}_i^{h'}$ is at least $1/\kappa^2$ times that of $\bar{x}_i^h$. A two-piece concave curve whose slope and cap are within factors $n\kappa^2$ and κ^2 dominates pointwise up to the factor $1/(n\kappa^2)$, proving the cover property; Lemma 11 finishes the proof. $\square$

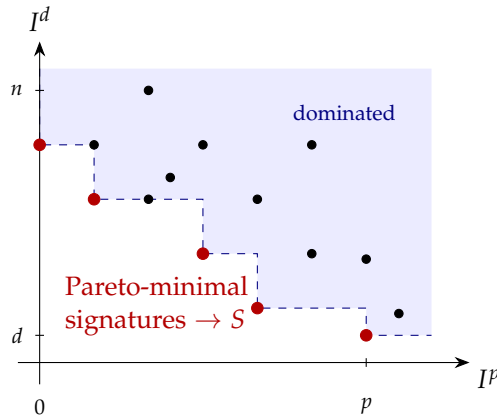


Figure 8: The Pareto-signature argument of Theorem 12: every scaled circuit $h \in \mathcal{C}'$ is plotted at its signature $(I^p(h), I^d(h)) \in \{0, \dots, p\} \times \{d, \dots, n\}$, where $p \leq m$ and $d \geq m + 1$. The Pareto-minimal signatures (red) form a strictly descending staircase — everything else lies in the shaded dominated region — so they take pairwise distinct I^p values and pairwise distinct I^d values, giving $|S| \leq \min\{p + 1, n - d + 1\} \leq \min\{m, n - m\} + 1$.

Consequences. Plugging Theorem 12 into Theorem 6: the TRUST-REGION IPM of Lecture 5 solves any LP with

$$O\left(n^{1.5} \min\{m, n - m\} \log(n\kappa_A)\right)$$

iterations — a bound depending on the constraint matrix alone. For bounded κ_A — network matrices ($\kappa = 1$, Lecture 3), and more generally Δ -modular integer matrices ($\kappa \leq \Delta$, by the Cramer’s rule exercise of Lecture 3) — this is a *strongly polynomial* iteration bound, recovering by interior point methods the territory of Tardos’s and Vavasis–Ye’s theorems from the Lecture 2 landscape [Tar86, VY96]. (One caveat: the theorem requires a well-centered starting point, and standard initialization schemes do not preserve κ ; making the initialization κ -safe requires additional ideas, see [DKN⁺24].) The companion bound of [DKN⁺24], $\text{SLC}_{1/(mn)}(x_i^m) = O(mn)$ whenever A has at most two nonzero entries per column, rests on the same concept of circuit covers — but not on Theorem 12: the circuit imbalance κ_A can be unbounded for generalized flow matrices, so the combinatorial construction of a small cover for the circuits of Exercise 4 is much more involved. This is the route to the strongly polynomial algorithm for such LPs mentioned in Lecture 2.

Looking ahead. Straight line complexity now stands as both floor and ceiling for path-following interior point methods: no such method can beat $\max_i \text{SLC}(x_i^m)$ many iterations (Proposition 5), instances exist where this is exponential (Theorem 8), and it is polynomial whenever the circuit imbalance is controlled (Theorem 12). What we have not yet done is exhibit the algorithm achieving the matching upper bound of Theorem 6. The next — and final — lecture builds it: the predictor-corrector interior point method, its classical $O(\sqrt{n} \log(1/\varepsilon))$ analysis, and the trust-region step that lets it travel down straight stretches of the central path at full speed. For further reading, see [ABGJ18] for the tropical lower bound, [ADL⁺22] for the SLC framework and the matching upper bound, and [DKN⁺24, ENV22] for the circuit imbalance theory.

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